

Stephen G. Ware, Ph.D.

<http://cs.uky.edu/~sgware> sgware@cs.uky.edu Last updated May 28, 2026

Biography

Stephen is an Associate Professor in the Department of Computer Science at the University of Kentucky, where he directs the Narrative Intelligence Lab and teaches courses on artificial intelligence and game development.

He uses AI to model interactive stories in virtual environments like video games and training simulations. Virtual reality experiences are more effective when the simulated characters adapt to accommodate the player's choices. Stephen works on strong story narrative planning systems (contrasted with emergent story systems) which quickly consider millions of hypothetical futures to find optimal stories given the experience designer's aesthetic and pedagogic goals.

His work has earned six best paper awards and been nominated for two more. Since 2014, he has participated in over \$3 million in sponsored federal, state, and local research, including a CAREER award from the U.S. National Science Foundation. He has published over 50 peer-reviewed papers. According to Google Scholar, his h-index is 17 and his work receives over 120 citations a year. He referees for the *IEEE Transactions on Games* journal and has served as an organizer or program committee member for top conferences in his field, including AIIDE, CoG, IVA, ICIDS, FDG, AAAI, and IJCAI.

Employment

Associate Professor with Tenure University of Kentucky Department of Computer Science	Summer 2023 to present
Consultant Disney Research Studios, Zürich	Summer 2022 to Spring 2023
Assistant Professor University of Kentucky Department of Computer Science	Fall 2019 to Summer 2023
Assistant Professor University of New Orleans Department of Computer Science	Fall 2014 to Summer 2019
Instructor, Research Assistant, and Teaching Assistant North Carolina State University Department of Computer Science	Fall 2008 to Spring 2014
Software Engineer Intern DAXCO Inc. Birmingham, AL, USA	Summer 2007

Education

Doctor of Philosophy in Computer Science, North Carolina State University Thesis: <i>A Plan-Based Model of Conflict for Narrative Reasoning and Generation</i>	June 2014
Master of Science in Computer Science, North Carolina State University GPA 4.0 / 4.0	May 2011
Bachelor of Science in Computer Science and Philosophy, Loyola University New Orleans Summa cum laude with university honors, GPA 4.0 / 4.0	May 2008

Awards

Best Paper: <i>Speeding Up Narrative Planning Using Fog of War Pruning</i> 21 st AAAI conference on Artificial Intelligence and Interactive Digital Entertainment	November 2025
Best Paper: <i>Language Models as Narrative Planning Heuristics</i> Foundations of Digital Games conference	April 2025
Best Student Paper: <i>A Model for Automating the Abstraction of Planning Problems in a Narrative Context</i> 20 th AAAI conference on Artificial Intelligence and Interactive Digital Entertainment	November 2024
Best Program Committee Member 19 th AAAI conference on Artificial Intelligence and Interactive Digital Entertainment	October 2023
Early Career Creativity, Research, and Scholarship Award Univeristy of New Orleans	January 2019
Best Program Committee Member 13 th AAAI conference on Artificial Intelligence and Interactive Digital Entertainment	October 2017
Faculty Distinguished Research Award University of New Orleans Honors Program	April 2017
Region 1 Postsecondary Teacher of the Year Louisiana Association of Computer Using Educators	December 2016
Best Student Paper: <i>Glaive: A State-Space Narrative Planner Supporting Intentionality and Conflict</i> 10 th AAAI conference on Artificial Intelligence and Interactive Digital Entertainment	October 2014
Best Paper: <i>Achieving the Illusion of Agency</i> International Conference on Interactive Digital Storytelling	July 2012
Best Student Paper on a Cognitive Science Topic: <i>Indexer: A Computational Model of the Event-Indexing Situation Model for Characterizing Narratives</i> Computational Models of Narrative workshop	May 2012
Outstanding Teaching Assistant North Carolina State University Graduate Student Association	March 2010
Dean's Fellowship North Carolina State University College of Engineering	August 2009
William T. Cotton Service Award Loyola Univeristy New Orleans	May 2008
Percy A. Roy S.J. Award for Highest Grade Point Average Loyola University New Orleans College of Humanities and Natural Sciences	May 2008
Outstanding Computer Science Major Loyola University New Orleans Department of Mathematics and Computer Science	May 2008
Guy Lemieux S.J. Award for Excellence in Philosophy Loyola University New Orleans Department of Philosophy	May 2008
Ignatian Scholarship Loyola University New Orleans	May 2004

Sponsored Research

About \$95,000 in internal grants omitted for brevity.

Large-Scale Evaluation of AI Collaborative Storytelling via the First Interactive Narrative Challenge (INCH), University of Kentucky Lighthouse Beacon Foundation Endowment
Principal Investigator / \$25,000 total

Supporting Players of Interactive Narrative Games by Recognizing Beliefs, Intentions, Memory, and Expectations, U.S. Army Research Lab
Principal Investigator / \$503,047 total

Inferring Domain Models to Help Analysts with Belief and Intention Recognition, Laboratory for Analytic Science, U.S. Department of Defense
Co-Principal Investigator / \$224,993 total / \$101,000 my share

Incentivization and Gamification Techniques to Promote Bedside Teaching During Patient Rounds and Reduce Neurophobia, American Board of Psychiatry and Neurology
Co-Investigator / \$100,000 total / \$50,000 my share

CAREER: Structured High-Agency Interactive Narratives for Virtual Environments, U.S. National Science Foundation
Principal Investigator / \$530,369 total

Visualizing and Assisting the Analytic Process, Laboratory for Analytic Science, U.S. Department of Defense
Co-Principal Investigator / \$182,453 total / \$91,227 my share

Hierarchical Abstract Process Modeling for Analyst Workflow, Laboratory for Analytic Science, U.S. Department of Defense
Co-Principal Investigator / \$182,453 total / \$91,227 my share

Explainable Interventions for Analyst Workflow, Laboratory for Analytic Science, U.S. Department of Defense
Co-Principal Investigator / \$182,438 total / \$91,219 my share

CHS: Small: Strong Story Narrative Planning for Authoring Proactive Intelligent Virtual Environments, U.S. National Science Foundation
Principal Investigator / \$493,256 total

Foreseeing, Recognizing, and Influencing Possible Futures Using Multi-Agent Planning Algorithms, Laboratory for Analytic Science, U.S. Department of Defense
Principal Investigator / \$80,543 total

CC Network Design: ARCHES (Advanced Research Computing in the Humanities Engineering and Sciences) Network at the University of New Orleans (equipment grant)*, U.S. National Science Foundation
Co-Principal Investigator / \$335,000 total

EAGER: Planning Believable Narratives by Modeling Agent Beliefs, U.S. National Science Foundation
Principal Investigator / \$156,969 total

CRII: CHS: Structuring Narratives in Interactive Virtual Environments Using Computational Models of Possible Worlds, U.S. National Science Foundation
Principal Investigator / \$138,436 total

Creating an Interdisciplinary Digital Media Laboratory (equipment grant), Louisiana May 2015 to May 2016
Board of Regents Enhancement Program
Principal Investigator / \$110,042 total

Publications

Refereed Journal Articles

9. R. Farrell and S. G. Ware, "Large language models as narrative planning search guides," *IEEE Transactions on Games*, vol. 17, no. 2, pp. 419–428, 2024
8. A. Shirvani et al., "Personality and emotion in strong-story narrative planning," *IEEE Transactions on Games*, vol. 15, no. 4, pp. 669–682, 2023
7. S. G. Ware et al., "Multi-agent narrative experience management as story graph pruning," *IEEE Transactions on Games*, vol. 15, no. 3, pp. 378–387, 2022
6. M. Siler and S. G. Ware, "Solution density and search strategy in narrative generation," *IEEE Transactions on Games*, vol. 14, no. 4, pp. 715–724, 2022
5. R. Farrell et al., "Manipulating narrative salience in interactive stories using Indexer's Pairwise Event Salience Hypothesis," *IEEE Transactions on Games*, vol. 12, no. 1, pp. 74–85, 2020
4. S. G. Ware and R. M. Young, "Intentionality and conflict in The Best Laid Plans interactive narrative virtual environment," *IEEE Transactions on Computational Intelligence and Artificial Intelligence in Games*, vol. 8, no. 4, pp. 402–411, 2015
3. B. Harrison et al., "A survey and analysis of techniques for player behavior prediction in Massively Multiplayer Online games," *IEEE Transactions on Emerging Topics in Computing Special Issue on MMO Technologies*, vol. 3, no. 2, pp. 260–274, 2014
2. S. G. Ware et al., "A computational model of narrative conflict at the fabula level," *IEEE Transactions on Computational Intelligence and Artificial Intelligence in Games*, vol. 6, no. 3, pp. 271–288, 2014
1. R. M. Young et al., "Plans and planning in narrative generation: A review of plan-based approaches to the generation of story, discourse and interactivity in narratives," *Sprache und Datenverarbeitung, Special Issue on Formal and Computational Models of Narrative*, vol. 37, no. 1-2, pp. 41–64, 2013

Refereed Conference Papers

33. L. Senanayake and S. G. Ware, "Speeding up narrative planning using Fog of War Pruning," in *Proceedings of the 21st AAAI conference on Artificial Intelligence and Interactive Digital Entertainment*, 2025, pp. 121–131, 24% acceptance rate, awarded Best Paper
32. M. Siler and S. G. Ware, "An answer set encoding for narrative planning with theory of mind," in *Proceedings of the 21st AAAI conference on Artificial Intelligence and Interactive Digital Entertainment*, 2025, pp. 132–141, 24% acceptance rate
31. F. N. Khan et al., "Detecting duplicate states is worth it in narrative planning with belief," in *Proceedings of the 21st AAAI conference on Artificial Intelligence and Interactive Digital Entertainment*, 2025, pp. 62–69, 24% acceptance rate
30. M. Siler et al., "Pareto-based narrative planning: Making NPCs more rational," in *Proceedings of the 21st AAAI conference on Artificial Intelligence and Interactive Digital Entertainment*, full paper presented as poster, 2025, pp. 337–346, 54% acceptance rate

29. M. Fisher et al., "Structure, agency, and improvisation in human-led digital interactive narrative exercises," in *Proceedings of the 21st AAAI conference on Artificial Intelligence and Interactive Digital Entertainment*, full paper presented as poster, 2025, pp. 237–246, 54% acceptance rate
28. G. Birchmeier and S. G. Ware, "Speeding up narrative planning with Causal Width Search and Pruning," in *Proceedings of the 21st AAAI conference on Artificial Intelligence and Interactive Digital Entertainment*, full paper presented as poster, 2025, pp. 196–205, 54% acceptance rate
27. L. Senanayake and S. G. Ware, "Language models as narrative planning heuristics," in *Proceedings of the 20th international conference on the Foundations of Digital Games*, 2025, 50% acceptance rate, awarded Best Paper
26. M. Fisher and S. G. Ware, "A model for automating the abstraction of planning problems in a narrative context," in *Proceedings of the 20th AAAI conference on Artificial Intelligence and Interactive Digital Entertainment*, 2024, pp. 35–45, 27% acceptance rate, awarded Best Student Paper
25. S. G. Ware et al., "Causal necessity as a narrative planning step cost function," in *Proceedings of the 19th AAAI conference on Artificial Intelligence and Interactive Digital Entertainment*, 2023, pp. 155–164, 29% acceptance rate
24. S. G. Ware and R. Farrell, "Salience as a narrative planning step cost function," in *Proceedings of the IEEE Conference on Games*, 2022, pp. 433–440, 44% acceptance rate
23. R. Farrell et al., "Salience vectors for measuring distance between stories," in *Proceedings of the 18th AAAI conference on Artificial Intelligence and Interactive Digital Entertainment*, full paper presented as poster, 2022, pp. 95–104, 62% acceptance rate
22. S. G. Ware and M. Siler, "Sabre: A narrative planner supporting intention and deep theory of mind," in *Proceedings of the 17th AAAI conference on Artificial Intelligence and Interactive Digital Entertainment*, 2021, pp. 99–106, 29% acceptance rate, nomination for Best Paper
21. M. Siler and S. G. Ware, "A good story is one in a million: Solution density in narrative generation problems," in *Proceedings of the 16th AAAI conference on Artificial Intelligence and Interactive Digital Entertainment*, 2020, pp. 123–129, 25% acceptance rate, nominated for Best Student Paper
20. R. Farrell and S. G. Ware, "Narrative planning for belief and intention recognition," in *Proceedings of the 16th AAAI conference on Artificial Intelligence and Interactive Digital Entertainment*, 2020, pp. 52–58, 25% acceptance rate
19. A. Shirvani and S. G. Ware, "A formalization of emotional planning for strong-story systems," in *Proceedings of the 16th AAAI conference on Artificial Intelligence and Interactive Digital Entertainment*, 2020, pp. 116–122, 25% acceptance rate
18. S. G. Ware et al., "Multi-agent narrative experience management as story graph pruning," in *Proceedings of the 15th AAAI conference on Artificial Intelligence and Interactive Digital Entertainment*, 2019, pp. 87–93, 25% acceptance rate
17. A. Shirvani and S. G. Ware, "A plan-based personality model for story characters," in *Proceedings of the 15th AAAI conference on Artificial Intelligence and Interactive Digital Entertainment*, full paper presented as poster, 2019, pp. 188–194, 49% acceptance rate
16. E. T. Garcia et al., "Measuring presence and performance in an intelligent virtual reality police use of force training simulation prototype," in *Proceedings of the 32nd AAAI International Florida Artificial Intelligence Research Society conference*, 2019, pp. 276–281, 52% acceptance rate
15. A. Shirvani et al., "Combining intentionality and belief: Revisiting believable character plans," in *Proceedings of the 14th AAAI conference on Artificial Intelligence and Interactive Digital Entertainment*, full paper presented as poster, 2018, pp. 222–228, 50% acceptance rate

14. A. Shirvani et al., “A possible worlds model of belief for state-space narrative planning,” in *Proceedings of the 13th AAAI conference on Artificial Intelligence and Interactive Digital Entertainment*, 2017, pp. 101–107, 25% acceptance rate
13. R. Farrell and S. G. Ware, “Influencing user choices in interactive narratives using Indexter’s Pairwise Event Salience Hypothesis,” in *Proceedings of the 13th AAAI conference on Artificial Intelligence and Interactive Digital Entertainment*, 2017, pp. 37–42, 25% acceptance rate
12. R. Farrell and S. G. Ware, “Causal link semantics for narrative planning using numeric fluents,” in *Proceedings of the 13th AAAI conference on Artificial Intelligence and Interactive Digital Entertainment*, full paper presented as poster, 2017, pp. 193–199, 50% acceptance rate
11. R. Farrell and S. G. Ware, “Fast and diverse narrative planning through novelty pruning,” in *Proceedings of the 12th AAAI conference on Artificial Intelligence and Interactive Digital Entertainment*, 2016, pp. 37–43, 28% acceptance rate
10. R. Farrell and S. G. Ware, “Predicting user choices in interactive narratives using Indexter’s Pairwise Event Salience Hypothesis,” in *Proceedings of the 9th International Conference on Interactive Digital Storytelling*, 2016, pp. 147–155, 36% acceptance rate
9. R. Farrell et al., “Asking hypothetical questions about stories using QUEST,” in *Proceedings of the 9th International Conference on Interactive Digital Storytelling*, 2016, pp. 136–146, 36% acceptance rate
8. C. Kives et al., “Evaluating the Pairwise Event Salience Hypothesis in Indexter,” in *Proceedings of the 11th AAAI conference on Artificial Intelligence and Interactive Digital Entertainment*, 2015, pp. 30–36, 28% acceptance rate
7. S. G. Ware and R. M. Young, “Glaive: A state-space narrative planner supporting intentionality and conflict,” in *Proceedings of the 10th AAAI international conference on Artificial Intelligence and Interactive Digital Entertainment*, 2014, pp. 80–86, 26% acceptance rate, awarded Best Student Paper
6. R. E. Cardona-Rivera et al., “Foreseeing meaningful choices,” in *Proceedings of the 10th AAAI conference on Artificial Intelligence and Interactive Digital Entertainment*, 2014, pp. 9–15, 26% acceptance rate
5. M. W. Fendt et al., “Achieving the illusion of agency,” in *Proceedings of the 5th International Conference on Interactive Digital Storytelling*, 2012, pp. 114–125, 29% acceptance rate, awarded Best Paper
4. S. G. Ware et al., “Four quantitative metrics describing narrative conflict,” in *Proceedings of the 5th International Conference on Interactive Digital Storytelling*, 2012, pp. 18–29, 29% acceptance rate
3. S. G. Ware and R. M. Young, “Validating a plan-based model of narrative conflict,” in *Proceedings of the 7th international conference on the Foundations of Digital Games*, 2012, pp. 220–227, 29% acceptance rate
2. S. G. Ware and R. M. Young, “CPOCL: A narrative planner supporting conflict,” in *Proceedings of the 7th AAAI conference on Artificial Intelligence and Interactive Digital Entertainment*, 2011, pp. 97–102, 35% acceptance rate
1. S. G. Ware and R. M. Young, “Modeling narrative conflict to generate interesting stories,” in *Proceedings of the 6th AAAI conference on Artificial Intelligence and Interactive Digital Entertainment*, full paper presented as poster, 2010, pp. 210–215, 33% acceptance rate

Refereed Workshop and Consortium Papers

12. M. Siler et al., “Designing a modular, scalable benchmark for narrative experience management,” in *Proceedings of the joint workshop on Experimental AI in Games and Intelligent Narrative Technologies at the 21st AAAI conference on Artificial Intelligence and Interactive Digital Entertainment*, 2025
11. M. Siler and S. G. Ware, “Structure, agency, and intent: Preliminary data collection,” in *Proceedings of the Interactive Narrative Technologies workshop at the 20th AAAI conference on Artificial Intelligence and Interactive Digital Entertainment*, 2024

10. M. Fisher et al., “Intelligent de-escalation training via emotion-inspired narrative planning,” in *Proceedings of the 13th Intelligent Narrative Technologies workshop at the 18th AAAI conference on Artificial Intelligence and Interactive Digital Entertainment*, 2022
9. S. G. Ware and M. Fisher, “Exploring regression-based narrative planning,” in *Proceedings of the 12th Intelligent Narrative Technologies workshop at the 16th AAAI conference on Artificial Intelligence and Interactive Digital Entertainment*, 2020
8. A. Shirvani and S. G. Ware, “Camelot: A modular customizable sandbox for visualizing interactive narratives,” in *Proceedings of the 12th Intelligent Narrative Technologies workshop at the 16th AAAI conference on Artificial Intelligence and Interactive Digital Entertainment*, 2020
7. A. Shirvani and S. G. Ware, “On automatically motivating story characters,” in *Proceedings of the Experimental AI in Games workshop at the 15th AAAI conference on Artificial Intelligence and Interactive Digital Entertainment*, 2019
6. S. G. Ware, “Mutual Implicit Question Answering for shared authorship: A pilot study on player expectations,” in *Proceedings of the 10th Intelligent Narrative Technologies workshop at the 13th AAAI conference on Artificial Intelligence and Interactive Digital Entertainment*, 2017, pp. 259–265
5. S. G. Ware, “The Intentional Fast-Forward narrative planner,” in *Proceedings of the 5th Intelligent Narrative Technologies workshop at the 8th AAAI conference on Artificial Intelligence and Interactive Digital Entertainment*, 2012, pp. 57–62
4. R. E. Cardona-Rivera et al., “Indexer: A computational model of the Event-Indexing Situation Model for characterizing narratives,” in *Proceedings of the 3rd workshop on Computational Models of Narrative*, 2012, pp. 34–43, awarded Best Student Paper on a Cognitive Science Topic
3. S. G. Ware et al., “Initial results for measuring four dimensions of narrative conflict,” in *Proceedings of the 4th Intelligent Narrative Technologies workshop at the 7th AAAI conference on Artificial Intelligence and Interactive Digital Entertainment*, 2011, pp. 115–122
2. S. G. Ware, “A computational model of narrative conflict,” in *Doctoral Consortium at the 6th international conference on the Foundations of Digital Games*, 2011, pp. 247–249
1. S. G. Ware and R. M. Young, “Rethinking traditional planning assumptions to facilitate narrative generation,” in *Proceedings of the AAAI Fall Symposium on Computational Models of Narrative*, 2010, pp. 71–72

Refereed Posters and Extended Abstracts

1. S. G. Ware and M. Siler, “The Sabre narrative planner: Multi-agent coordination with intentions and beliefs,” in *Proceedings of the 20th International Conference on Autonomous Agents and Multiagent Systems*, 2021, pp. 1698–1700

Refereed Book Chapters

2. S. G. Ware, “An introduction to Graph Theory,” in *Practical Graph Mining with R*, N. F. Samatova et al., Eds., CRC Press, 2013, pp. 9–26. [Online]. Available: <http://www.crcpress.com/product/isbn/9781439860847>
1. B. Harrison et al., “Frequent Subgraph Mining,” in *Practical Graph Mining with R*, N. F. Samatova et al., Eds., CRC Press, 2013, pp. 181–221. [Online]. Available: <http://www.crcpress.com/product/isbn/9781439860847>

Refereed Demonstrations

2. B. Samuel et al., “Playable experiences at AIIDE 2018,” in *Proceedings of the 14th AAAI conference on Artificial Intelligence and Interactive Digital Entertainment*, 2018, pp. 275–280
1. N. R. Sturtevant et al., “Playable experiences at AIIDE 2014,” in *Proceedings of the 10th AAAI conference on Artificial Intelligence and Interactive Digital Entertainment*, 2014, pp. 203–209

Non-Refereed Publications

4. S. G. Ware and R. Sanghrajka, “Workshop report: Intelligent Narrative Technologies: The 16th AAAI conference on Artificial Intelligence and Interactive Digital Entertainment,” in *Interactive AI Magazine*, 2021
3. S. G. Ware et al., “Interactive narrative planning in The Best Laid Plans,” in *Proceedings of the 25th AAAI conference on Artificial Intelligence, Virtual Agent Demonstrations*, 2015, pp. 4313–4314
2. O. Gown et al., “Reports on the 2012 AIIDE workshops,” *AI Magazine*, vol. 34, no. 1, pp. 90–92, 2012
1. S. G. Ware, *Crossed swords and broken hearts: A computational model of narrative conflict*, North Carolina State University Graduate Research Symposium, 2012

Keynotes and Invited Talks

<i>Innovations in Procedural Narrative</i> AI Summit, Game Developers Conference	March 2022
<i>Learning from Mis-Spun Tales</i> Immersive Stories Workshop, Virginia Tech Center for Human-Computer Interaction, Blacksburg, VA, USA	March 2021
<i>Learning from Mis-Spun Tales</i> Disney Research Studios, Zürich, Switzerland	April 2021

Panels

<i>Benchmarking Interactive Narrative Systems</i> , Panelist 12 th International Conference on Interactive Digital Storytelling, Snowbird, UT, USA	November 2019
<i>Narrative Intelligence in Interactive Storytelling</i> , Panelist 1 st NarraScope Conference, Massachusetts Institute of Technology, Boston, MA, USA	June 2019
<i>Expert Panel</i> , Panelist 5 th International Conference on Interactive Digital Storytelling, Technological Park, San Sebastián, Spain	November 2012
<i>The Near Future of Intelligent Narrative Technologies</i> , Moderator 5 th workshop on Intelligent Narrative Technologies, Stanford University, Palo Alto, CA, USA	October 2012
<i>Two Cultures: Crossing the Divide</i> , Panelist Collaborations: Humanities and Technology Festival, Duke University, Durham, NC, USA	February 2012

Teaching

University of Kentucky

- CS 660: Topics in Artificial Intelligence (AI for Interactive Narrative)
- Spring 2023: 6 graduate
 - Spring 2021: 5 graduate
- CS 665: Planning Algorithms
- Spring 2026: 8 graduate

- Spring 2025: 8 graduate (taught as CS 660: Topics in Artificial Intelligence)
 - Spring 2024: 10 graduate (taught as CS 660: Topics in Artificial Intelligence)
 - Spring 2022: 8 graduate (taught as CS 660: Topics in Artificial Intelligence)
 - Spring 2020: 6 graduate, 1 undergraduate (taught as CS 660/585: Topics in Artificial Intelligence)
- CS 463G: Introduction to Artificial Intelligence
- Spring 2026: 6 graduate, 40 undergraduate
 - Spring 2024: 60 undergraduate
 - Spring 2023: 65 undergraduate
 - Spring 2022: 54 undergraduate
 - Fall 2021: 55 undergraduate
 - Spring 2021: 45 undergraduate
- CS 410G: Introduction to Game Development
- Fall 2024: 40 undergraduate
 - Fall 2023: 40 undergraduate
 - Fall 2022: 56 undergraduate
 - Fall 2020: 34 undergraduate (taught as CS 485G: Special Topics)
 - Fall 2019: 55 undergraduate (taught as CS 485G: Special Topics)
- CS 217: Introduction to Data Structures
- Spring 2025: 28 undergraduate

University of New Orleans

- CSCI 6645: Planning Algorithms in Artificial Intelligence
- Fall 2017: 5 graduate
 - Fall 2016: 7 graduate
 - Fall 2015: 14 graduate (taught as CSCI 6990: Special Topics)
- CSCI 4525 / 5525: Introduction to Artificial Intelligence
- Spring 2019: 12 undergraduate, 9 graduate
 - Spring 2018: 18 undergraduate, 7 graduate
 - Spring 2017: 11 undergraduate, 6 graduate
 - Spring 2016: 17 undergraduate, 3 graduate
 - Spring 2015: 19 undergraduate, 14 graduate
- CSCI 4675 / 5675: Advanced Game Development
- Spring 2018: 5 undergraduate, 4 graduate
 - Spring 2017: 9 undergraduate
 - Spring 2016: 7 undergraduate, 3 graduate
- CSCI 4670 / 5670: Fundamentals of Game Development
- Fall 2018: 17 undergraduate, 2 graduate
 - Fall 2017: 20 undergraduate, 7 graduate
 - Fall 2016: 13 undergraduate, 2 graduate
 - Fall 2015: 10 undergraduate, 10 graduate
 - Fall 2014: 11 undergraduate, 2 graduate
- CSCI 1583: Software Design and Development I
- Spring 2019: 14 undergraduate
 - Fall 2018: 32 undergraduate

North Carolina State University

- CSC 316: Data Structures for Computer Scientists
- Summer 2013: 21 undergraduate
- CSC 216: Programming Concepts – Java
- Summer 2009: 12 undergraduate

Service

Funding Panels and Reviews

Dutch Research Council The Hague, The Netherlands	December 2020
University of Costa Rica San José, Costa Rica	October 2019
National Sciences and Engineering Research Council of Canada Ottawa, Canada	January 2019
U.S. National Science Foundation, CISE Directorate Arlington, VA, USA	May 2015

Journal Referee

<i>IEEE Transactions on Games</i> (previously <i>IEEE Transactions on Computational Intelligence and AI in Games</i>)	Since October 2013
<i>Autonomous Agents and Multi-Agent Systems</i>	Since December 2019

Conference and Workshop Organization

Local Arrangements Chair 20 th AAAI conference on Artificial Intelligence and Interactive Digital Entertainment University of Kentucky, Lexington KY, USA	2024
General Chair 18 th AAAI conference on Artificial Intelligence and Interactive Digital Entertainment California State Polytechnic University, Pomona CA, USA	2022
Program Chair 17 th AAAI conference on Artificial Intelligence and Interactive Digital Entertainment Online due to COVID-19 pandemic	2021
Organizer 12 th workshop on Intelligent Narrative Technologies at 16 th AAAI conference on Artificial Intelligence and Interactive Digital Entertainment Online due to COVID-19 pandemic	2020
Organizer CamJam: Using the Camelot Virtual Environment workshop at the 15 th AAAI conference on Artificial Intelligence and Interactive Digital Entertainment Georgia Institute of Technology, Atlanta GA, USA	2019
Doctoral Consortium Chair 14 th AAAI conference on Artificial Intelligence and Interactive Digital Entertainment University of Alberta, Edmonton, Canada	2018
Organizer 7 th workshop on Computational Models of Narrative at the Digital Humanities conference Kraków, Poland	2016

5th workshop on Intelligent Narrative Technologies at the 8th AAAI conference on Artificial Intelligence and Interactive Digital Entertainment
Stanford University, Palo Alto, CA, USA

Program Committees

AAAI conference on Artificial Intelligence and Interactive Digital Entertainment (AIIDE)

- 21st conference, University of Alberta, Edmonton, Canada, November 2025
- 20th conference, University of Kentucky, Lexington, KY, USA, November 2024
- 19th conference, University of Utah, Salk Lake City, UT, USA, October 2023
- 18th conference, California State Polytechnic Univ., Pomona, CA, USA, October 2022
- 17th conference, online due to COVID-19 pandemic, October 2021
- 16th conference, online due to COVID-19 pandemic, October 2020
- 15th conference, Georgia Institute of Technology, Atlanta, GA, USA, October 2019
- 14th conference, University of Alberta, Edmonton, Canada 2018
- 13th conference, Snowbird, UT, USA, October 2017
- 12th conference, Burlingame, CA, USA, October 2016
- 11th conference, Univ. of California Santa Cruz, Santa Cruz, CA, USA, November 2015
- 10th conference, North Carolina State University, Raleigh, NC, USA, October 2014

IEEE Conference on Games (CoG)

- 1st conference, London, United Kingdom, August, 2019

ACM conference on Intelligent Virtual Agents (IVA)

- 15th conference, Delft University of Technology, Delft, The Netherlands, August 2015

International Conference on Interactive Digital Storytelling (ICIDS)

- 13th conference, Bournemouth University, Bournemouth, United Kingdom, November 2020
- 12th conference, Snowbird, UT, USA, November 2019
- 10th conference, Funchal, Madeira, Portugal, November 2017
- 9th conference, Institute for Creative Technologies, Los Angeles, CA, USA, November 2016
- 5th conference, Technological Park, San Sebastián, Spain, November 2012

International conference on the Foundations of Digital Games (FDG)

- 9th conference, Royal Caribbean Liberty of the Seas, April 2014
- 7th conference, Raleigh, North Carolina, USA, May 2012

AAAI conference on Artificial Intelligence (AAAI)

- 36th conference, Vancouver, Canada, February 2022
- 35th conference, online due to COVID-19 pandemic, February 2021
- 32nd conference, New Orleans, LA, USA, February 2018

International Joint Conference on Artificial Intelligence (IJCAI)

- 34th conference, Special Track on AI, Arts and Creativity, Montreal, CA, August, 2025
- 33rd conference, Special Track on AI, Arts, and Creativity, Jeju, S. Korea, August, 2024
- 29th conference, Yokohama, Japan, July 2020
- 26th conference, Melbourne, Australia, August 2017

Experimental AI in Games workshop (EXAG)

- 12th workshop, University of Alberta, Edmonton, Alberta, Canada, November 2025
- 11th workshop, University of Kentucky, Lexington, Kentucky, USA, November 2024
- 9th workshop, California State Polytechnic University, Pomona, CA, USA, October 2022
- 6th workshop, Georgia Institute of Technology, Atlanta, GA, USA, October 2019

Intelligent Narrative Technologies workshop (INT)

- 14th workshop, University of Kentucky, Lexington, KY, USA, November 2024
- 13th workshop, California State Polytechnic Univ., Pomona, CA, USA, October 2022
- 12th workshop, online due to COVID-19 pandemic, October 2020

- 11th workshop, held jointly with Workshop on Intelligent Cinematography and Editing, University of Alberta, Edmonton, Alberta, Canada, November 2018
 - 10th workshop, Snowbird, UT, USA, October 2017
 - 9th workshop, special track of the 9th International Conference on Interactive Digital Storytelling, Los Angeles, CA, USA, November 2016
 - 8th workshop, University of California Santa Cruz, Santa Cruz, CA, USA, November 2015
 - 7th workshop, University of Wisconsin-Milwaukee, Milwaukee WI, USA, June 2014
 - 6th workshop, Northeastern University, Boston MA, USA, October 2013
 - 5th workshop, Stanford University, Palo Alto CA, USA, October 2012
- Computational Models of Narrative workshop (CMN)
- 9th workshop, Universidad Complutense de Madrid, Spain, June 2026
 - 8th workshop, University of Geneva, Geneva, Switzerland, May 2025
 - 7th workshop, Kraków, Poland, July 2016
 - 4th workshop, University of Hamburg, Berlin, Germany, August 2013

University of Kentucky

Chair, Media and Outreach Committee Department of Computer Science	Fall 2024 to Present
Faculty Advisor, Wildcat Games Studio Office of Student Organizations and Activities	Spring 2024 to Present
Member, Committee on Higher Degrees Department of Computer Science	Fall 2021 to Present
Member, Hiring Committee Department of Computer Science	Fall 2021 to Spring 2025
Member, Media and Outreach Committee Department of Computer Science	Summer 2019 to Fall 2024

University of New Orleans

Faculty Advisor, International Game Developers Association, Student Chapter Office of Student Organizations	Fall 2015 to Spring 2019
Member, Undergraduate Studies Committee Department of Computer Science	Fall 2015 to Spring 2019

Research Supervision

Doctor of Philosophy, as Advisor

6. Gage Birchmeier, University of Kentucky (degree in progress)	Fall 2024 to Present
5. Lasantha Senanayake, University of Kentucky (degree in progress)	Spring 2023 to Present
4. Mira Fisher, University of Kentucky (degree in progress)	Summer 2020 to Present
3. Molly Siler, University of Kentucky (degree in progress)	Spring 2020 to Present
2. Alireza Shirvani, University of Kentucky Thesis: <i>Personality and Emotion for Virtual Characters in Strong-Story Narrative Planning</i>	Spring 2017 to Summer 2021
1. Rachelyn Farrell, University of Kentucky Thesis: <i>Don't Give Me That Story! Human-Centered Narrative Generation and Summarization of Possible Stories</i>	Spring 2015 to Fall 2022

Doctor of Philosophy, as Committee Member

12. Jonathan Sekela, University of Kentucky (degree in progress) Spring 2025 to Present
11. Nicholas Fluty, University of Kentucky (degree in progress) Spring 2025 to Present
10. Muhammad Haider, University of Kentucky (degree in progress) Fall 2024 to Present
9. Motasem Obeidat, University of Kentucky (degree in progress) Summer 2024 to Present
8. Luke Elmore, University of Kentucky (degree in progress) Spring 2023 to Present
7. Adib Mosharrof, University of Kentucky Spring 2023 to Spring 2026
Thesis: *Advancing Task-Oriented Dialog Systems: Scalability, Generalization, and Evaluation*
6. Elinor Rubin-McGregor, University of Kentucky (degree in progress) Fall 2022 to Present
5. Nathan Arnold, University of Kentucky Fall 2021 to Fall 2024
Thesis: *Tiered Coalition Formation Game Variants, Stability, and Simulation*
4. Anton Vinogradov, University of Kentucky Spring 2021 to Summer 2024
Thesis: *Recovering From And Accommodating For Player Preference Shifts In A Player Modeled Experience Management Environment*
3. Giulio Mori, University of Reykjavik Spring 2020 to Summer 2023
Thesis: *Platform for Decoupling Experience Managers and Environments*
2. Tasmia Tasrin, University of Kentucky Spring 2020 to Present
Thesis: *Augmenting Machine Learning Technique Through Natural Language*
1. Md. Sultan Al Nahian, University of Kentucky Fall 2019 to Fall 2023
Thesis: *Practical AI Value Alignment Using Stories*

Master of Science, as Advisor

5. Jacob Finerty, University of Kentucky Fall 2022
Project: *Implementing Iterated Width Search into the Sabre Narrative Planner*
4. Dustin Peabody, University of New Orleans Spring 2016 to Spring 2018
Thesis: *Detecting Metagame Shifts in League of Legends Using Unsupervised Learning*
3. Edward T. Garcia, University of New Orleans Fall 2015 to Summer 2019
Thesis: *Multi-Agent Narrative Experience Management as Story Graph Pruning*
2. Dharmesh Desai, University of New Orleans Fall 2015 to Spring 2017
Thesis: *Measuring Presence in a Police Use of Force Simulation*
1. Rachelyn Farrell, University of New Orleans Summer 2015 to Spring 2017
Thesis: *Predicting User Choices in Interactive Narratives using Indexer's Pairwise Event Salience Hypothesis*

Master of Science, as Committee Member

13. Jeffrey Nance, University of Kentucky Spring 2026
Project: *Automated Team Generation Using Multi-Agent Reinforcement Learning*
12. Riley Sheridan, University of Kentucky Spring 2026
Project: *One-Shot Prompted Extraction of Initial Game State Information from Super Smash Bros. Melee Using ChatGPT-5.4*
11. Sandesh Lamichhane, University of Kentucky Spring 2024
Project: *ABC to TAB*
10. Ethan Coots, University of Kentucky Spring 2024
Project: *Tractor Front-End Loader Image Classification using CNNs*
9. David Bosomworth, University of Kentucky Fall 2023
Project: *Code Generation Using Interpretable Machine Learning Policies*
8. Vincent Liu, University of Kentucky Spring 2023
Thesis: *Language Controlled Interfaces for Game Environments*
7. Jonathan Moore, Air Force Institute of Technology Spring 2023
Thesis: *Designing an Adaptable Game Framework to Teach Joint All-Domain Concepts in a Variety of Training Environments*
6. Ryan Lindsey, University of Kentucky Fall 2022
Project: *Novel 3D Environment for Reinforcement Learning*

- | | |
|--|-------------|
| 5. Michael Probst, University of Kentucky | Fall 2021 |
| Thesis: <i>Maintaining Game Event Sequences in Procedurally Generated Levels</i> | |
| 4. Ryan Bowles, University of Kentucky | Spring 2021 |
| Project: <i>Penumbra Passage: A Single Player Digital Board Game</i> | |
| 3. Devi Kavya Sri Regati, University of Kentucky | Spring 2020 |
| Project: <i>Implementing Preference Reasoning over Answer Set Optimization Problems</i> | |
| 2. Dheeraj Betha, University of Kentucky | Fall 2019 |
| Project: <i>Using Omiotis for Software Tracing</i> | |
| 1. Shyla Clark, University of New Orleans | Spring 2019 |
| Thesis: <i>Remote Monitoring of Cherry Wetness Using a Leaf Wetness Sensor and Wireless Sensor Network</i> | |

Bachelor of Science, as Thesis Advisor or Independent Study Supervisor, etc.

- | | |
|--|----------------------------|
| 41. Calvin Higdon, University of Kentucky | Fall 2024 |
| 40. Aaron Hall, University of Kentucky | Spring 2024 |
| 39. Sydney Chapman, University of Kentucky | Spring 2024 |
| 38. Jackson Cotthoff, University of Kentucky | Spring 2024 |
| 37. Matthew Melton, University of Kentucky | Spring 2023 |
| 36. Travis Nilest, University of Kentucky | Spring 2023 |
| 35. Joseph Alverson, University of Kentucky | Spring 2023 |
| Thesis: <i>Desktop Drummer: Creating a Rhythm-Based Video Game with a Custom Controller</i> | |
| 34. Zachary Coleman, University of Kentucky | Spring 2022 |
| 33. Alexa Griffin, University of Kentucky | Spring 2022 |
| 32. Matthew Mitchell, University of Kentucky | Fall 2021 |
| 31. Gareth Walker, University of Kentucky | Fall 2021 |
| 30. Ethan Hall, University of Kentucky | Fall 2021 to Spring 2022 |
| 29. Alexander Dingus, University of Kentucky | Spring 2021 |
| 28. Evan Shepherd, University of Kentucky | Spring 2021 |
| 27. John Colfer, University of Kentucky | Spring 2020 |
| 26. Zachary Foster, University of Kentucky | Spring 2020 |
| 25. Jacob Hayden, University of Kentucky | Spring 2020 |
| 24. Mac McNeerney, University of Kentucky | Spring 2020 |
| 23. Jean-Paul Jeunesse, University of New Orleans | Fall 2018 to Spring 2019 |
| Thesis: <i>Measuring Interactive Narrative Quality with Experience Management as Story Graph Pruning</i> | |
| 22. Rishav Rajendra, University of New Orleans | Spring 2018 to Fall 2018 |
| 21. Lee Lagarde, University of New Orleans | Spring 2018 to Fall 2018 |
| 20. Ted Mader, University of New Orleans | Fall 2016 to Spring 2017 |
| Thesis: <i>Integrating Virtual Reality with Use-of-Force Training Simulations</i> | |
| 19. Nicholas Martin, University of New Orleans | Spring 2017 |
| 18. Nishan Rayamajhee, University of New Orleans | Spring 2017 |
| 17. Ashim Sitoula, University of New Orleans | Spring 2016 |
| 16. Pujan Pokhrel, University of New Orleans | Spring 2016 |
| 15. Scott Robertson, University of New Orleans | Spring 2016 |
| 14. Hung Le, University of New Orleans | Spring 2016 |
| 13. Abhishek Sapkota, University of New Orleans | Fall 2015 to Spring 2016 |
| 12. Rodrigo Rodrigues do Carmo, University of New Orleans | Spring 2015 to Summer 2015 |
| 11. Maurice Robert III, University of New Orleans | Summer 2015 |
| 10. Thiago Vieira, University of New Orleans | Summer 2015 |
| 9. Gabriel Miranda, University of New Orleans | Summer 2015 |
| 8. Gabriel Queiroz, University of New Orleans | Spring 2015 |
| 7. Christopher Toups, University of New Orleans | Spring 2015 |
| 6. Christian Stith, North Carolina State University | Fall 2013 |
| 5. Phillip Wright, North Carolina State University | Fall 2013 |
| 4. Eric Lang, North Carolina State University | Fall 2012 |

3. Zack Litzsinger, North Carolina State University
2. Evan Kochuk, North Carolina State University
1. Courtney Harrison, North Carolina State University

Fall 2012
Spring 2011
Spring 2011

Professional Organizations

- AAAI: Association for the Advancement of Artificial Intelligence (member #53757)
- ACM: Association for Computing Machinery (member #2211285)
- IEEE: Institute of Electrical and Electronics Engineers (member #92209981)
- IGDA: International Game Developers Association (member #22066812)
- ARDIN: Association for Research on Digital Interactive Narratives

Publicity

- UK Researcher Enhances Police De-Escalation Training with Immersive AI, VR Simulations* 6 March 2025
UKNow: University of Kentucky News
<https://uknow.uky.edu/research/uk-researcher-enhances-police-de-escalation-training-immersive-ai-vr-simulations>
- Stephen Ware's NSF-funded Research Trains Police in De-escalation* 12 July 2022
UKNow: University of Kentucky News
<https://uknow.uky.edu/research/stephen-ware-s-nsf-funded-research-trains-police-de-escalation>
- Stephen Ware Receives NSF CAREER Award* 28 April 2022
University of Kentucky College of Engineering News
<https://engr.uky.edu/news/stephen-ware-receives-nsf-career-award>
- UNO's Stephen Ware Recognized as Post-Secondary Teacher of the Year by Louisiana Association of Computer Using Educators* 16 November 2016
University of New Orleans Campus News
https://www.uno.edu/campus-news/2016/UNOs_Stephen_Ware_Recognized_as_Post_Secondary_Teacher_of_the_Year_by_Louisiana_Association_of_Computer_Using_Educators.aspx
- Computer Science Professor Wins \$157,000 NSF Grant to Study Narrative Intelligence* 26 July 2016
University of New Orleans Campus News
https://www.uno.edu/campus-news/2016/Computer_Science_Professor_Wins_157k_NSF_Grant_to_Study_Narrative_Intelligence.aspx
- Game On: UNO's Video Game Development Concentration Simulates Real World Experience* 31 May 2016
University of New Orleans Campus News
<https://www.uno.edu/campus-news/2016/Game-On-UNO-Video-Game-Development-Concentration-Simulates-Real-World-Experience.aspx>
- University of New Orleans to Get New Digital Media Lab* 13 May 2015
NOLA.com
https://www.nola.com/news/education/university-of-new-orleans-to-get-new-digital-media-lab/article_4ff31368-9a7c-5cd3-8417-3b1e5adef7de.html
- UNO Professor Teaches Computers The Human Art Of Storytelling* 26 February 2015
All Things Considered, WWNO National Public Radio, New Orleans, LA, USA
<http://wwno.org/post/uno-professor-teaches-computers-human-art-storytelling>
- Guest* 12 February 2015
Think Tank with Garland Robinette, WWL Radio, New Orleans, LA, USA
<http://media.wwl.com/a/101898840/2-12-12pm-garland-uno-artificial-intelligence.htm>

UNO Professor Wins National Science Foundation Grant for Artificial Intelligence Research
NOLA.com

9 February 2015

https://www.nola.com/news/education/uno-professor-wins-national-science-foundation-grant-for-artificial-intelligence-research/article_05a1aec2-1657-551e-9ebb-0edccf14231c.html

Miscellaneous

- Erdős Number: 3 (via Judy Goldsmith)